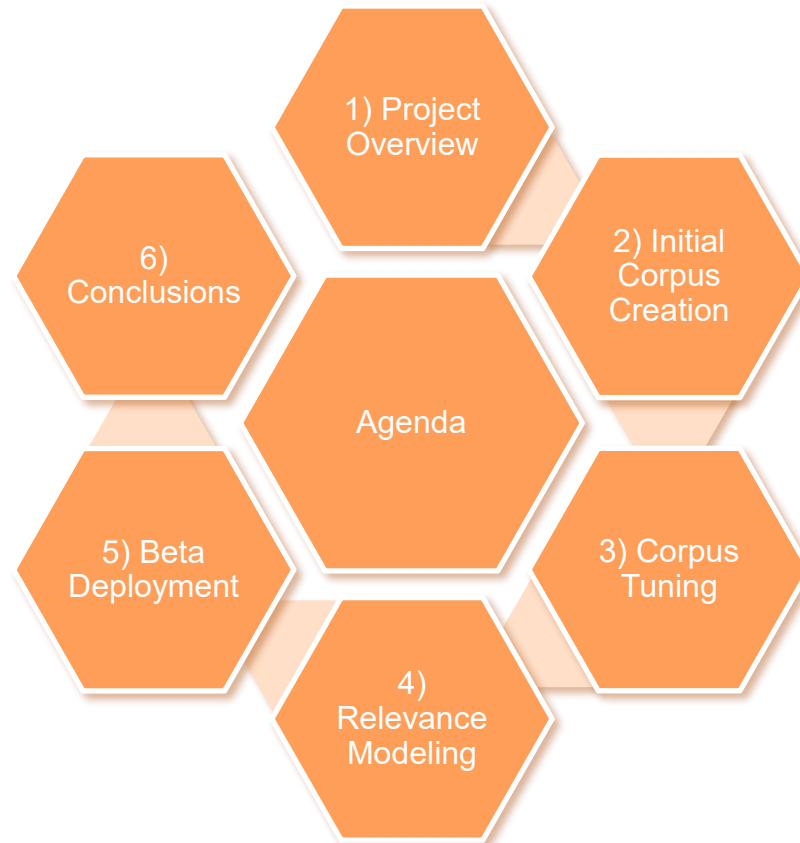


GrantMinded: Grant Identification Prototype

Sarah Gothard



Project Overview: The Issue

Nonprofits at structural disadvantage in finding grant funding for many reasons

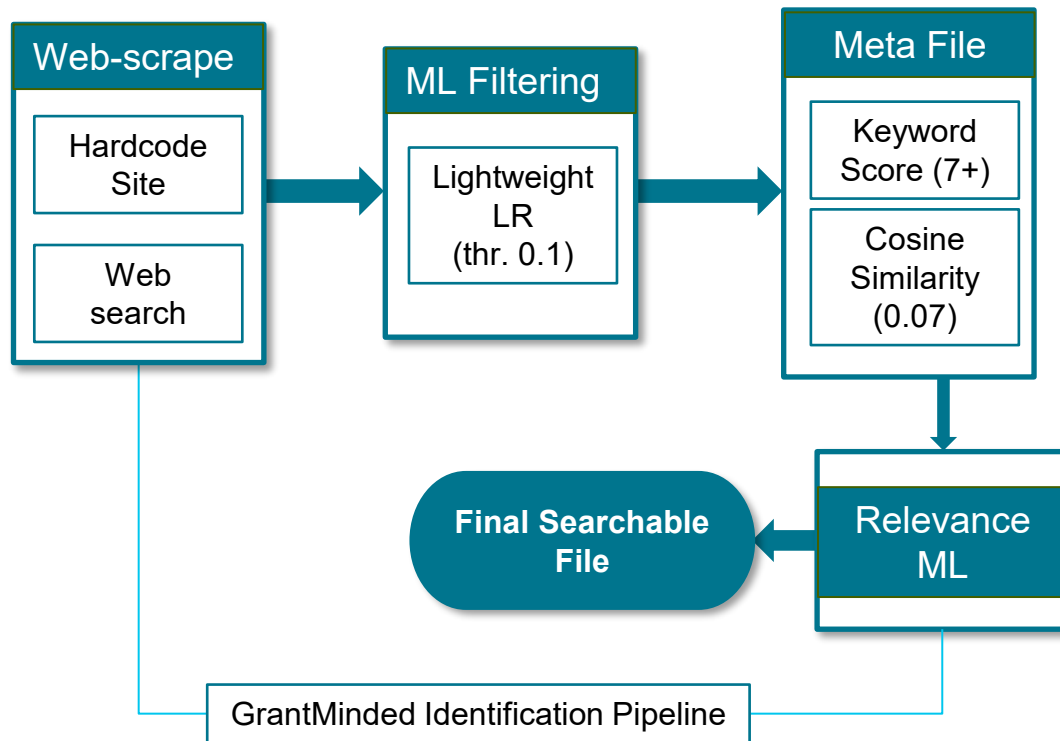
- Discovery is time-consuming
- The landscape is highly fragmented
- Heterogeneous information quality and language
- Mission-specific opportunities narrower than most funder interests
- Paid grant portals are costly, putting small nonprofits at a disadvantage
- 2 million+ nonprofits in USA: Highly competitive landscape



Project Overview: Solution

GrantMinded

- **Feasibility Experiment:** application is a prototype system designed to test the feasibility of automating grant discovery for small to mid-sized nonprofits.
- Tuned to I-CONNECT Foundation's mission



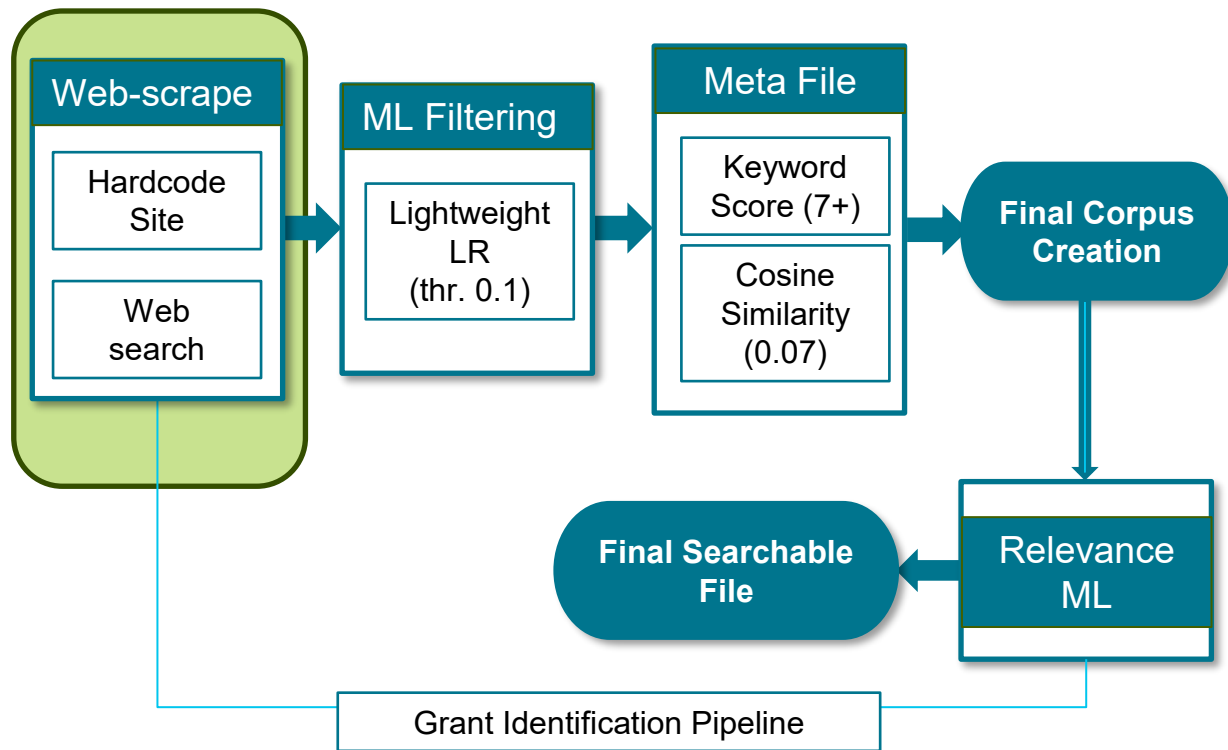
Project Overview: I-CONNECT Baseline

I-CONNECT Baseline (year prior)

- Part of staff dedicated time
- Uses a combination of solicitation, outreach, and web searchers to identify funding
- Costs of strategies are 10% of budget
- 3 grants relevant to mission



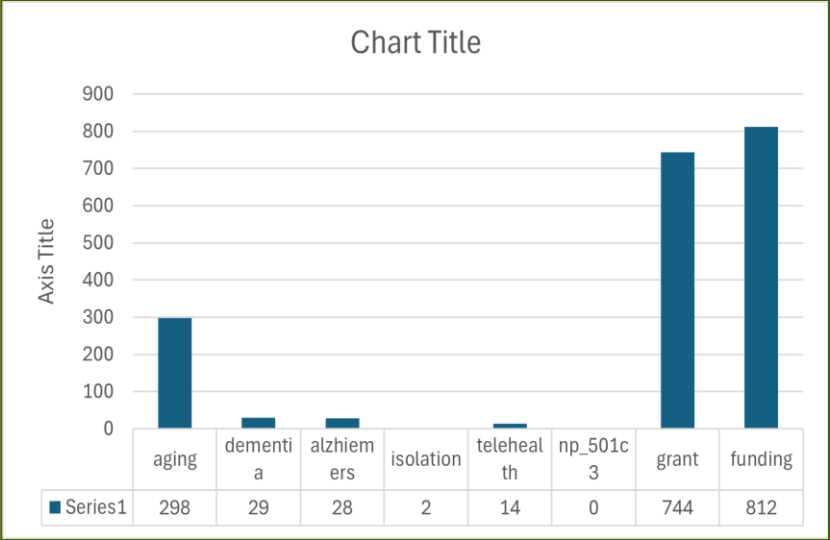
Corpus Pipeline Iteration: Stage 0



- Targeted Scraping: Crawls specific known grant portals and foundation sites.
- General Crawl: Expands to broader web searches,
- 1000 documents

Corpus Pipeline Iteration: Stage 0

Metric	Mean	SD	Min	Max	Max Aval.
Keywords	1.9	1.0	0.0	6.0	8.0
Keyword Score	6.6	3.4	0.0	18.0	24.0
Cosign Sim	0.0	0.0	0.0	0.3	1.0



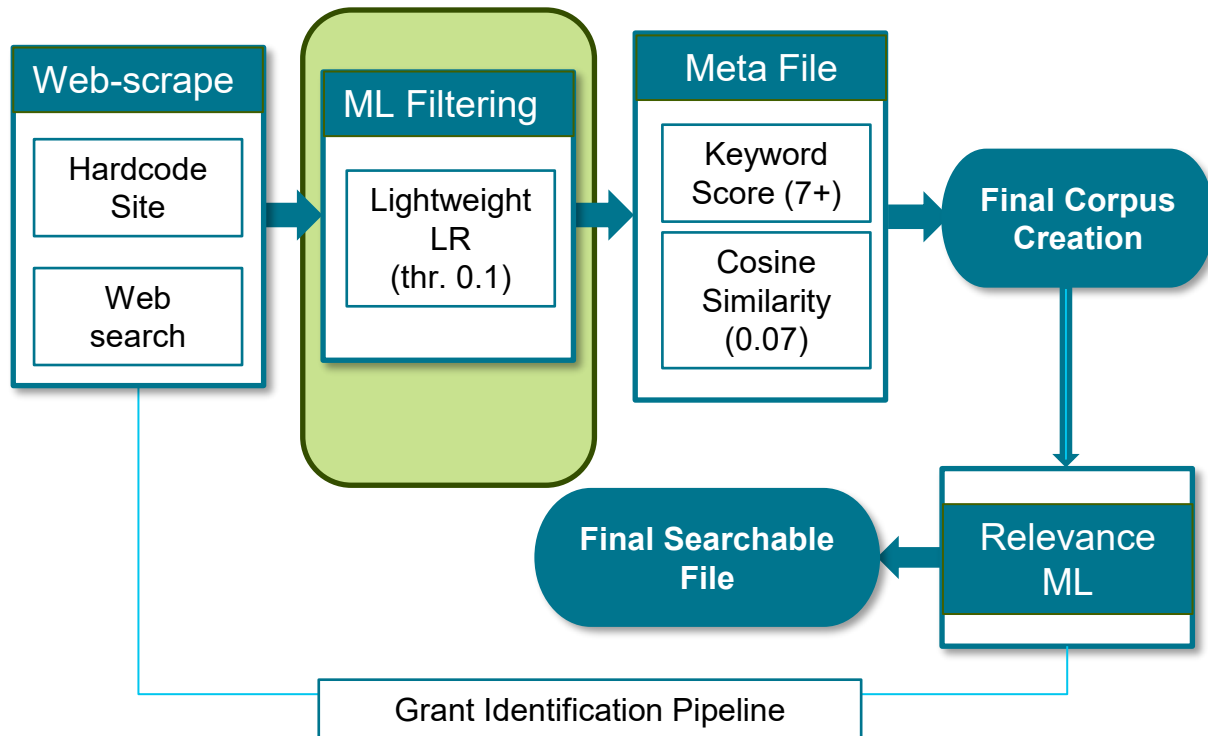
Corpus Generation Pipeline: Content

Relevance	Is Grant	Not a Grant	Unsure	Grand Total
Not Relevant	60	775	88	923
Relevant	13			13
Duplicate			64	64
Total	73	775	152	1000

Content Snippet	Relevance	Is Grant?
Making an impact Local impact Sustainable finance Supporting economic opportunity Environmental sustainability Find resources How we amplify the power of sports to drive progress, celebrate achievement and support communities Working here Explore Careers Investors Profile Quarterly earnings Events & presentations Regulatory & other filings Fixed income Shareholder info Press releases Annual reports & proxy	0	grant
Alzheimer's Foundation of America Upcoming Grant Information >Our Bi-Annual membership grants deliver funding for programs and services that help further s mission of providing support, service	1	grant

Corpus Pipeline Iteration: Stage 1

- Grant Identification was a challenge
- Modeled 2 ML
- Chose best performing to filter



Corpus Pipeline Iteration: “Is Grant” Split

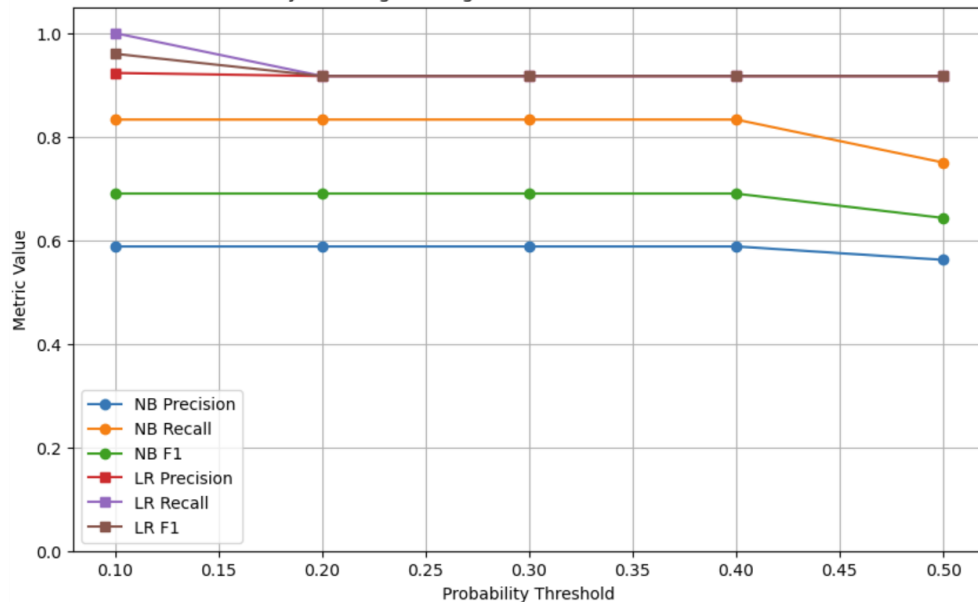
Initial Corpus

Initial Corpus	
Training data Is grant: 60 rows: 224	Test Data Is grant: 13 rows: 789

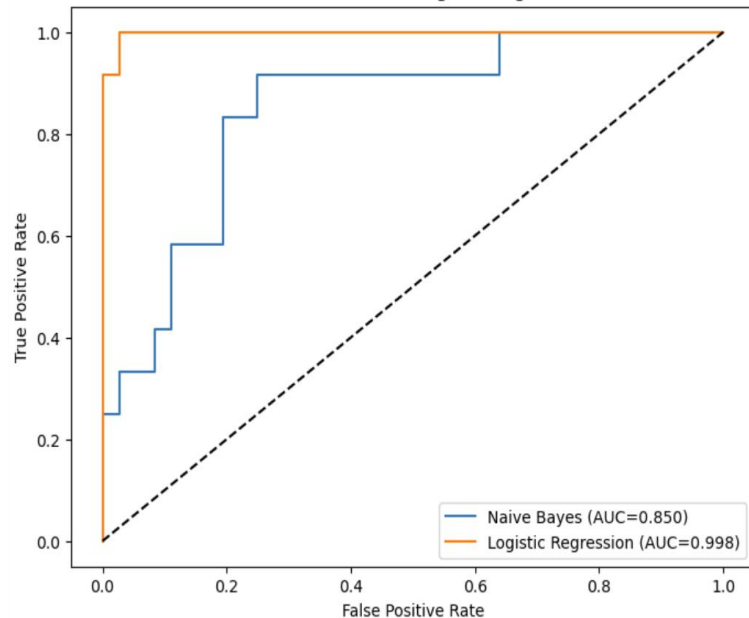
- Rare-Event Challenge: only 73 total true grants in 1k. Using full data set would bury noise
- More balanced training set – Training set had 224 total documents (60 grants/ 164 non grants) Still imbalanced but smaller so less noise
- Model Comparison: Evaluated Naïve Bayes vs. Logistic Regression, tuning thresholds
- comparing recall and F1 to select the most reliable classifier.

Corpus Pipeline Iteration: ML Reivew

Naive Bayes vs Logistic Regression Performance Across Thresholds

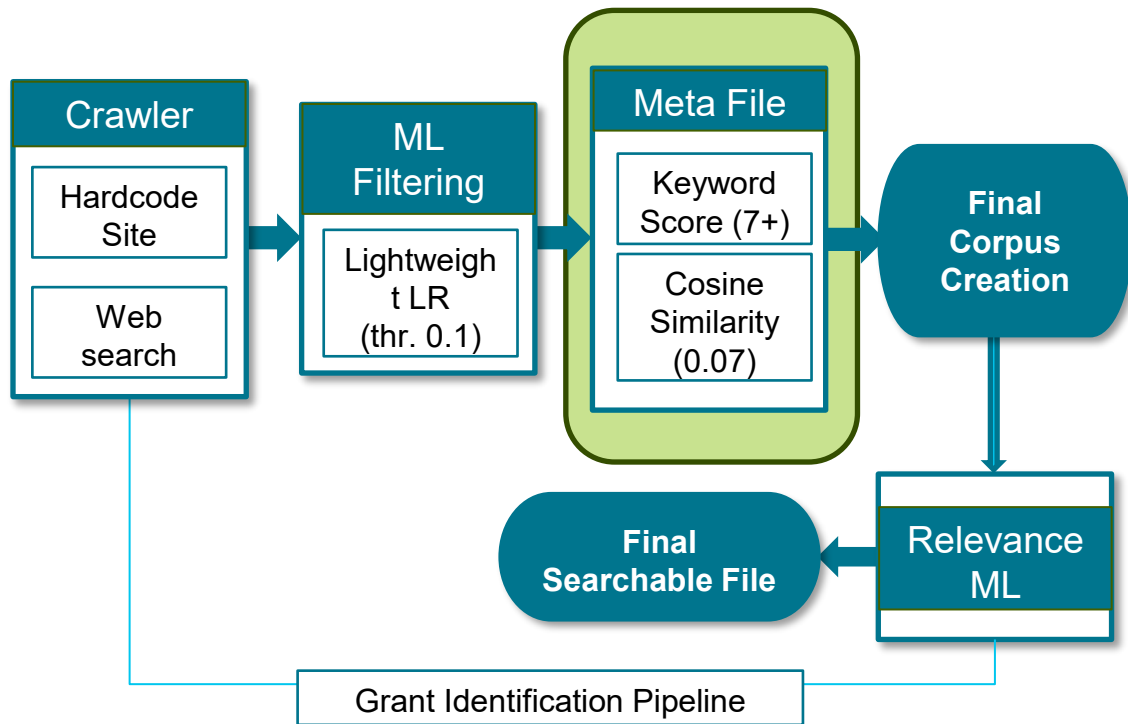


ROC Curves: NB vs Logistic Regression



Corpus Generation Pipeline: Stage 2

- Using same initial unfiltered corpus performed empirical review of numeric filtering metrics
- Keyword Search/Cosign Smiilarity



Corpus Generation Pipeline: Reivew

Keyword Score

- Significantly related ($p = 0.016$).
- Threshold: ≥ 7 captures most grants items.

Cosine Similarity

- Significantly related ($p < 0.0001$).
- Good separation: 0.10 vs. 0.04;
- Threshold: ≥ 0.07 retains ~90% of grants.

Both features are statistically significant and improve early-stage filtering.

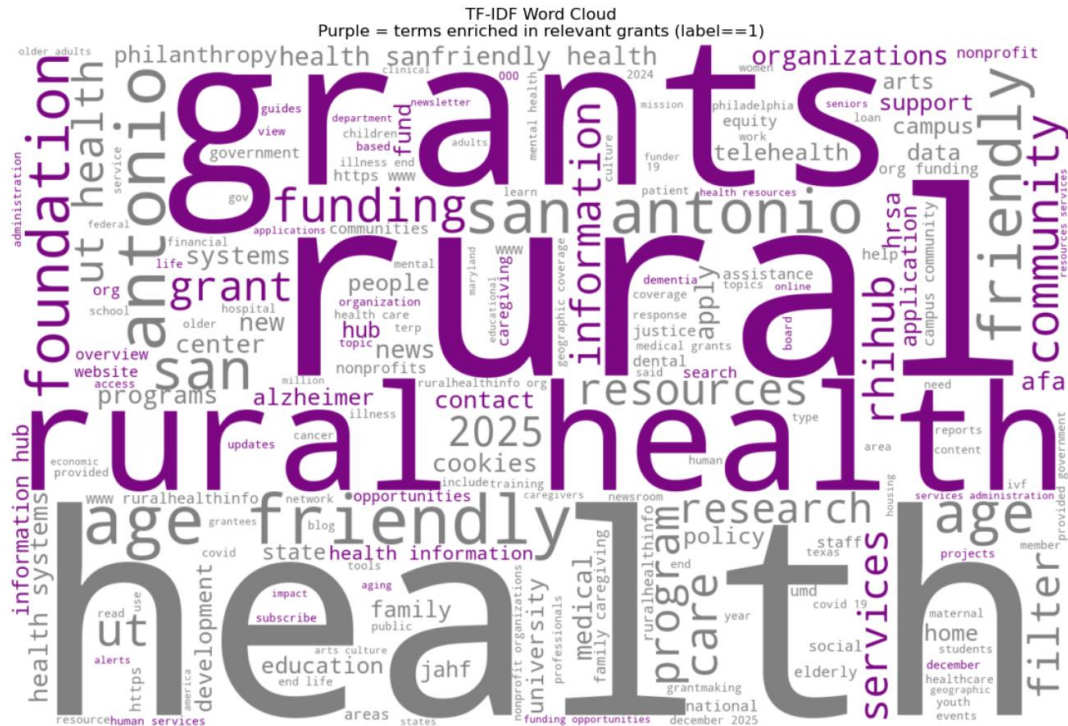
Metric	Mean	SD	P-Value
Keyword			0.016
<i>Not Relevant</i>	6.68	3.46	
<i>Relevant</i>	9.02	3.54	
Cosine Sim.			< 0.0001
<i>Not Relevant</i>	0.04	0.03	
<i>Relevant</i>	0.1	0.06	

Corpus Pipeline Iteration: Final Corpus

Metric	Overall	Not Relevant	Relevant
N	1013	985	28
Total Tokens	1,149,718	1,125,526	24,192
Vocabulary Size	22,345	22,186	2,690
Doc Length (mean)	1135.0	1142.7	864.0
Keyword Score (mean(sd))	6.0(3.2)	5.9 (3.1)	8.2 (2.6)
Cosine Sim. Score (mean(sd))	0.1 (0.04)	0.1 (0.04)	0.1 (0.1)

- Final corpus: 1,013 documents
- 3 major versions with 18 refactoring cycles.
- 28 relevant grants.
2x better than init.
9x better than baseline

- TfidfVectorizer tokenized
- Removed English stopwords
- TF-IDF downweight common words
- Included unigrams + bigrams (ngram_range = 1,2)
- Top 5000 features limit
- Rare terms kept if top 5000



Relevance Modeling: Split

Initial Corpus

Training data relevant: 20 rows: 725	Test Data subjects: 8 rows: 288

Split preserves the natural imbalance in the real grant landscape.

Train set: 725 pages total

- 20 relevant
- 705 not relevant

Test set: 288 pages total

- 8 relevant
- 280 not relevant

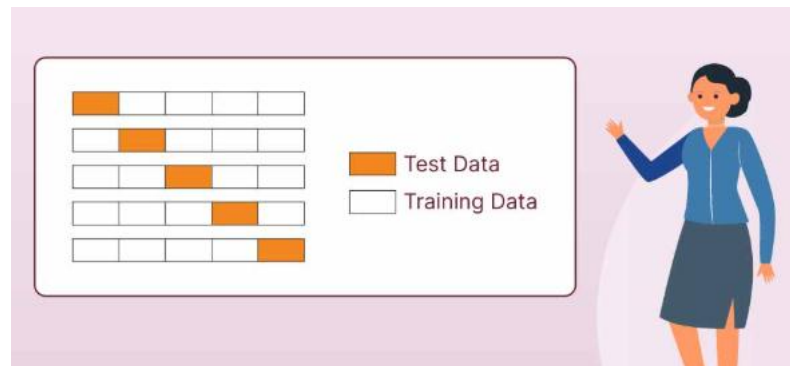
Relevance Modeling: Model Options

Performed Model Fitting on:

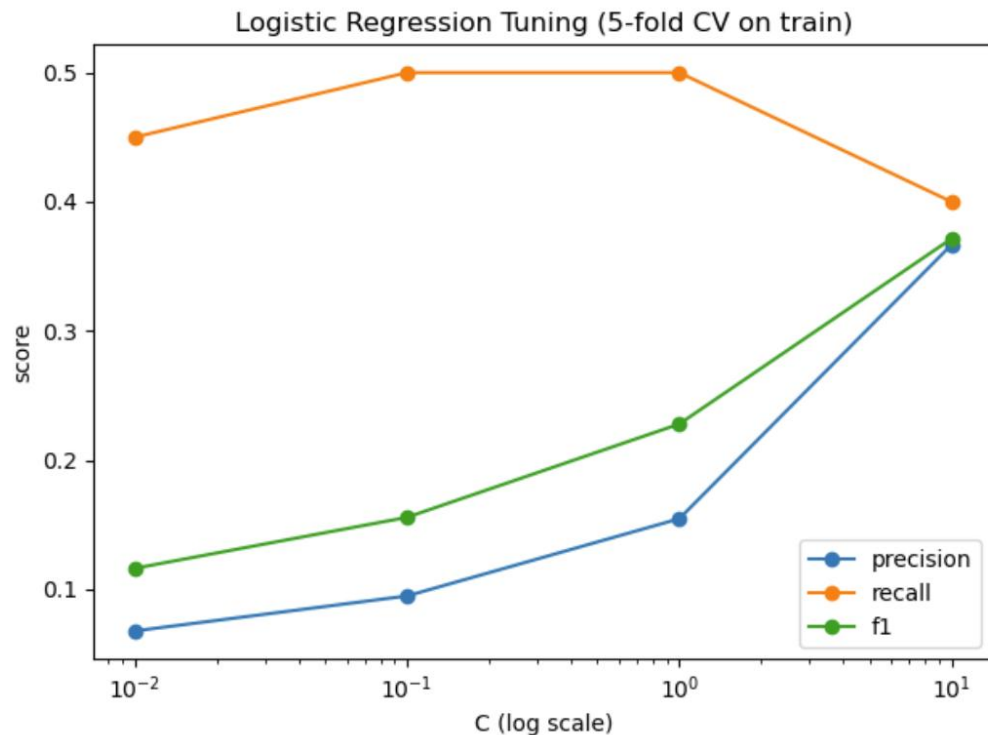
- Naïve Bayes
- Logistic Regression
- SVM

Relevance Modeling: K-Folds

- Used 5-fold stratified cross-validation to evaluate each model
- Rare relevant pages proportionate
- Stable metrics
- Trained and validated 5 times
- Precision, recall, and F1 are averaged for a more reliable estimate



Relevance Modeling: Logistic Regression



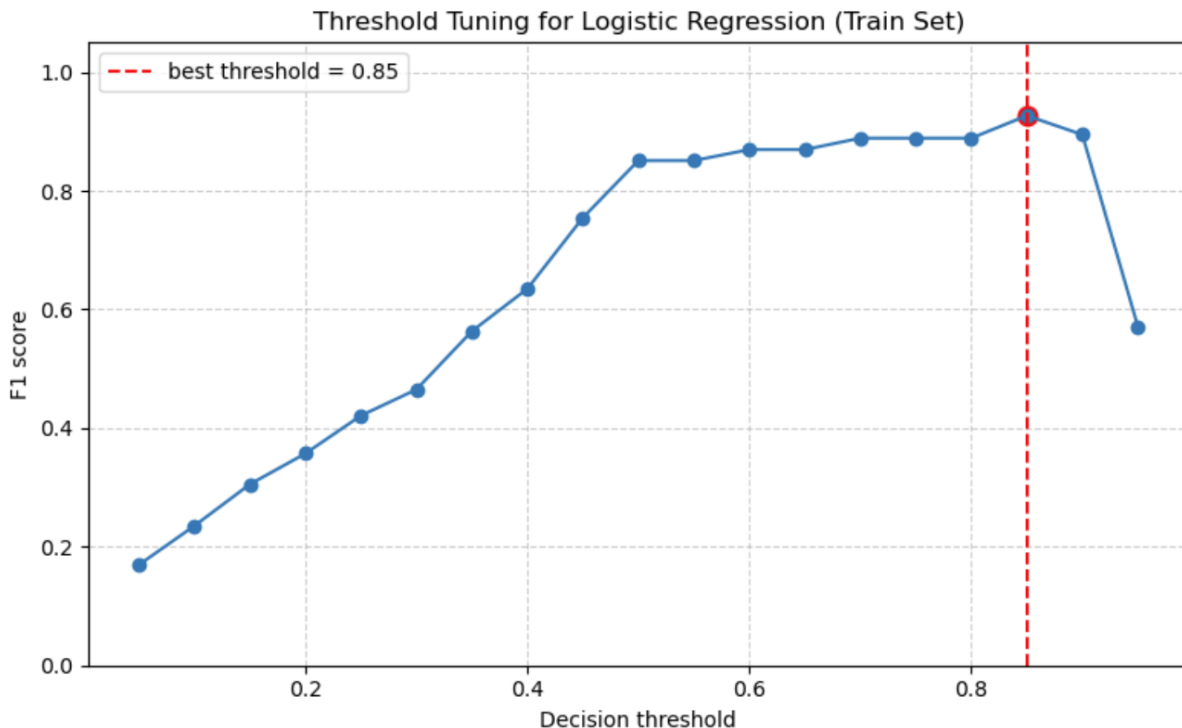
Training Tuning

- Rare relevant pages proportionate
- Stable metrics
- Trained and validated 5 times
- Precision, recall, and F1 are averaged for a more reliable estimate

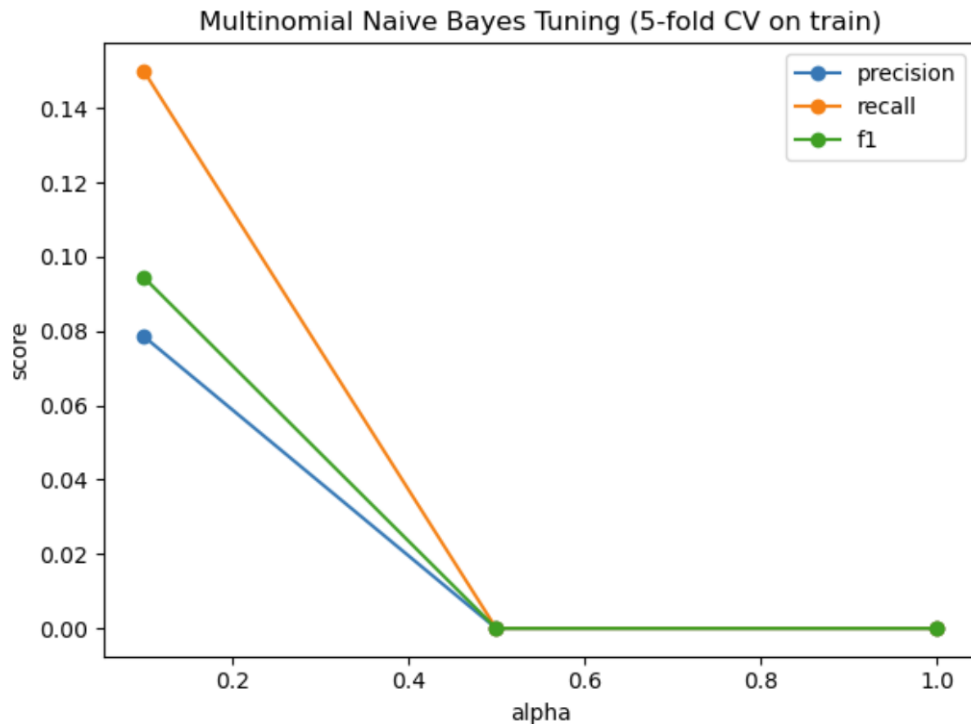
Relevance Modeling: Logistic Regression

Training Threshold Tuning

- Evaluate thresholds from **0.05 to 0.95**
- Optimized based on **F1 score**
- **Best threshold = 0.85**, producing the strongest F1 (0.927) on the training data



Relevance Modeling: Naïve Bayes

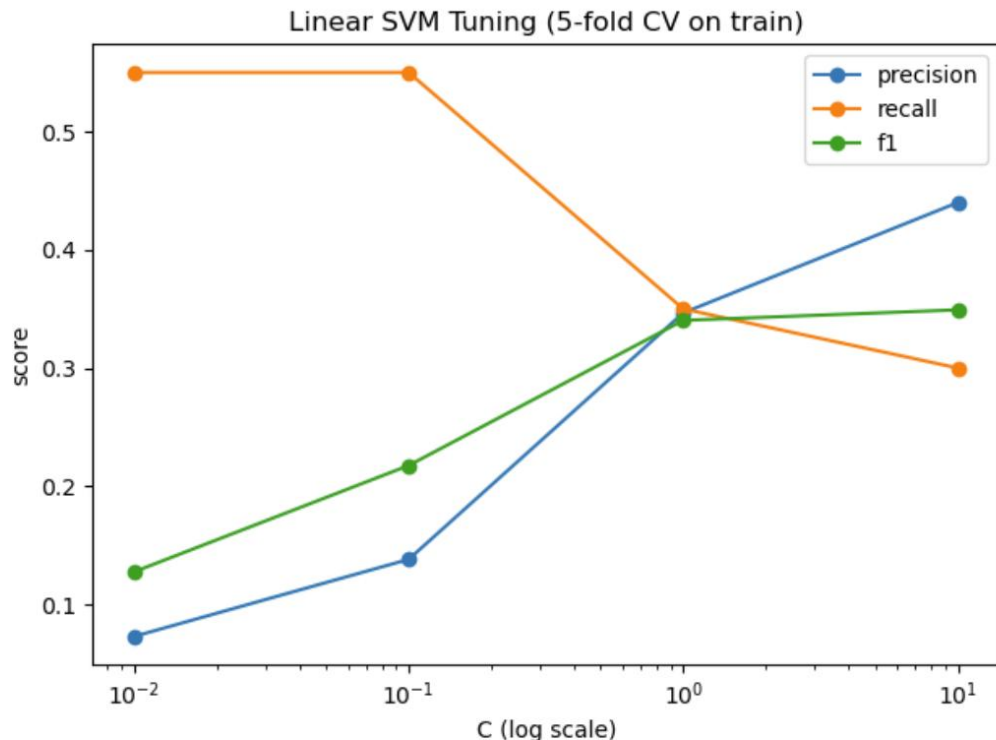


Smoothing value tuning

- **alpha = 0.1, 0.5, 1.0** using 5-fold CV
- Only **alpha = 0.1** produced non-zero precision/recall

NB struggled due to extreme rarity and high sparsity of relevant pages

Relevance Modeling: SVM



- $C = 0.01, 0.1, 1, 10$ | 5-fold
- Best at $C = 10$
- `Class_weight='balanced'` so rare relevant more influence

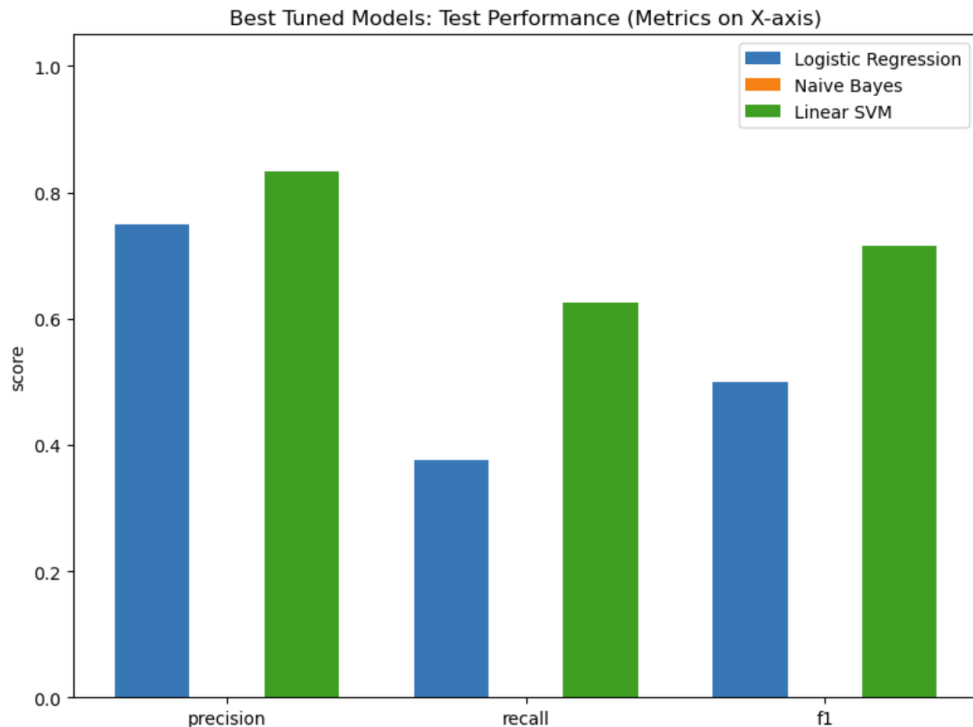
Relevance Modeling: Final Model

- **Linear SVM with C = 10** achieved the strongest performance on the final test set
 - Precision: **0.833**
 - Recall: **0.625**
 - F1: **0.714**
- Outperformed by a large margin

Output for Pipeline

`svm_vectorizer`: TF-IDF model used to transform all new pages

`svm_clf`: trained SVM relevance classifier for mission alignment



GrantMinded Demonstration 1 Results

- 3 Sites Crawled:
 - 0 grants identified
- 100% Correct



GrantMinded Demonstration 2 Results

Demo pipeline = 16 high-confidence pages

Manual review: **13 relevant (81.25%)**

Original labeled corpus= **2.8%** relevant

Final pipeline demo = ~ **30x enrichment** in relevant hits

In practical terms:

- Before: ~**36 pages** to find 1 relevant
- Now: ~**1–2 pages** to find 1 relevant

Relevant Status	Count	% of Total
Not Relevant	3	0.1875
Relevant	13	0.8125
Grand Total	16	



Conclusions

Major Findings

- ***Significant potential to reduce staff time and cost in grant identification***
- Pipeline behaved correctly across both demos
- Demonstrates clear feasibility and value of the multi-stage pipeline

Limitations

Relevance model must be retrained per nonprofit because mission focus differs

Future Work

- Build an application that automatically retrains for new missions
- Pilot at I-CONNECT, including cost/time savings analysis
- Additional tuning for stricter or more exploratory crawling mode

Citation

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